

Mehjabin Shirin

Ph.D. Scholar in Optometry and Vision Sciences | University of Hyderabad, India

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Academic Profile

Doctoral researcher in Optometry and Vision Sciences with interests in colour perception, face perception, visual psychophysics, eye-movement behaviour, retinal imaging, OCT-based biomarkers, cognitive ageing, early cognitive decline, Alzheimer's disease, and dementia biomarker research.

Research Interests

Colour and Visual Cognition: Colour Perception and Colour Cognition; Visual Adaptation; Visual Attention; Psychophysics; Computational Vision Science.

Face and Social Perception: Face Perception and Facial Attractiveness; Eye-Movement Behaviour; Eye Tracking; Psychophysiology.

Retinal and Neuro-Ophthalmic Research: Retinal Imaging and OCT; Retinal Biomarkers of Alzheimer's Disease; Neuro-Ophthalmology; Clinical Vision Science.

Cognitive Ageing and Dementia: Early Cognitive Decline; Dementia Biomarkers; Cognitive Ageing.

Education

Ph.D. in Optometry and Vision Sciences, University of Hyderabad | 2025 - Present

Advisor: Dr. Shiva Ram Male

Research focus: Visual perceptual, psychophysical and retinal biomarkers associated with early Alzheimer's disease and dementia in the Indian population.

Integrated Master of Optometry, University of Hyderabad | 2019 - 2025

Final grade: 83.41%; CGPA 8.78/10

Relevant coursework: Neuro-ophthalmology, clinical research methodology, advanced statistics, binocular vision, retinal diseases, ocular epidemiology, occupational vision.

Research and Clinical Experience

Research Assistant - DAC Brain Health Initiative | December 2025 - Present

- Contributing to multisite research on cognitive assessment, visual biomarkers, and harmonised data collection workflows.
- Supporting documentation and visual assessment workflows for Alzheimer's disease and brain-health research.

Research Assistant - Alzheimer's Biomarker Project, IIIT Hyderabad | March 2025 - May 2025

- Supported psychophysical testing, eye-tracking protocols, physiological signal analysis, and research data processing.

Research Associate - Face Recognition and Skin Tone Perception Project | May 2024 - May 2025

- Designed and implemented experimental protocols related to face perception, visual attention, and skin-tone perception.

Clinical Intern - LV Prasad Eye Institute, Visakhapatnam | 2023 - 2024

- Completed clinical training in comprehensive eye care, binocular vision evaluation, visual function assessment, and community outreach.

Publications

Male SR, Gandhi R, **Shirin M**, Govindahari V, Mondal S, Waheed F. Does a System for Verification of the Cleaning Process for Intraocular Surgical Instruments Offer a Path to Improved Quality of Care? *Investigative Ophthalmology & Visual Science*. 2025;66(8):2242.

Conference Presentations

Oral Presentation

- International Colour Vision Society Meeting 2026, Brighton, United Kingdom, August 14-18, 2026 - oral presentation / talk.

Poster Presentation

- Vision Sciences Society Meeting 2026, Florida, USA - accepted poster. Attendance was not possible due to visa-related constraints.
- Beauty and Change Conference 2024, Turin, Italy - poster presentation. Attendance was not possible.
- Chromotherapy in Mental Health: A Scoping Review, School of Life Sciences, University of Hyderabad, 2024 - poster presented.

E-Poster Presentation

- World Society of Paediatric Ophthalmologists and Strabismologists 2026 - e-poster presentation. Attendance was not possible.

Awards & Recognition

Research Awards: ICVS Student Travel Award; Laura Bassi Foundation Publication Support Award (2024); Laura Bassi Foundation Publication Support Award (2025).

Competitive Selections: Selected for Elsevier Vision Research Travel Award (VSS 2026). Unable to utilize due to visa-related inability to attend the conference.

Professional Affiliations: DAC Brain Health Initiative; Alzheimer's Association.

Technical Skills

Programming and analysis: Python, MATLAB, PsychoPy, statistical analysis, data visualisation, research data management.

Vision science methods: Colour vision assessment, psychophysical task design, contrast sensitivity, visual adaptation, visual attention experiments.

Physiological monitoring: Eye tracking, pupillometry, HRV, EDA, multimodal signal processing.

Clinical research: Participant recruitment, informed consent, clinical assessment workflows, community screening, ethics documentation.